

Hybrid Quantum-Classical Hierarchical Actor-Critic for Sum-Rate Maximization in IRS-Assisted Grouped RSMA Systems

Thinh Duong Tan Hung and Wonjong Noh

Abstract—Integrated satellite-terrestrial networks are impaired by building blockage of the direct path and by the worst-user bottleneck that a shared common stream imposes. We consider an IRS-assisted grouped rate-splitting multiple access downlink and jointly optimize four coupled decision variables, namely IRS-user link assignment, phase-shift configuration, transmit power allocation and common-rate distribution, under discrete phase quantization, blockage and imperfect channel state information. Their strict sequential dependency motivates a hybrid quantum-classical hierarchical actor-critic framework in which a variational quantum circuit resolves the combinatorial assignment while three classical actors govern the remainder. A direct state encoding would grow the register with network size, so an autoencoder holds it at eight qubits across both scales. Against a per-state alternating-optimization solver the framework gains roughly 30% on the QoS-penalized objective at five users and one IRS, narrowing to seed-variance parity at ten users and two IRSs. The quantum actor matches an equivalent classical network within seed variance at 6 to $9 \times$ fewer inference parameters, and every learned policy issues three rate-kernel invocations per decision regardless of user count, against 482 and 896 for the solver. The advantage accordingly resides in parameter efficiency and amortized inference rather than in achievable rate.

Index Terms—Hierarchical actor-critic (HAC), integrated satellite-terrestrial networks (ISTN), intelligent reflecting surface (IRS), quality of service (QoS), rate-splitting multiple access (RSMA), sum-rate maximization, variational quantum circuit (VQC).

I. INTRODUCTION

A. Background

Integrated satellite-terrestrial networks (ISTNs) extend coverage far beyond the reach of fixed ground infrastructure, and falling launch costs have made constellation-scale deployment practical [1]–[3]. The orbital link is nonetheless physically fragile. The extended path incurs substantial free-space path loss, the atmosphere adds rain attenuation and tropospheric scintillation [4], and in densely built environments buildings shadow the direct satellite-to-user link, depriving the affected terminal of a reliable primary propagation path.

What the geometry leaves behind is addressed by two techniques, the first of which confronts the worst-user bottleneck, whereby the terminals in the most adverse conditions constrain the aggregate rate of the entire served population [5]. Rate-splitting multiple access (RSMA) relaxes that bottleneck by decomposing each message into a common component, decoded by every receiver in the designated set and then removed by successive interference cancellation [6], [7], and a private component decoded independently, a grouped extension bounding the decodable common rate by the weakest member of a subgroup rather than by the globally worst user [8]. Rate splitting cannot by itself remedy an obstructed

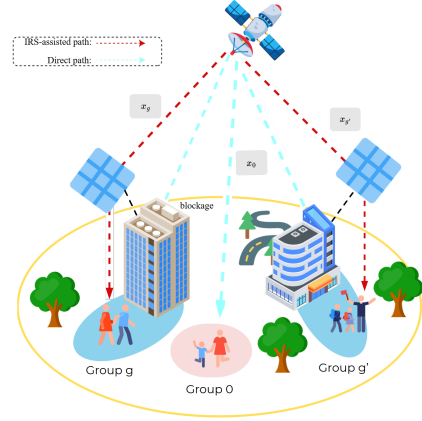


Fig. 1: Proposed IRS-assisted grouped RSMA system in an ISTN downlink. The satellite simultaneously serves $G + 1$ groups: Group 0 users with unobstructed satellite links receive the direct signal x_0 , while Group g and Group g' users blocked by buildings are served via dedicated IRS panels carrying reflected signals x_g and $x_{g'}$. Each IRS serves its assigned group through the phase-shift configuration Φ_m , chosen so that its reflected contributions combine coherently at those users.

path, for which an intelligent reflecting surface (IRS) [9] supplies an alternative route. Its passive elements each impose an independently controllable phase shift, quantized in practice to a few discrete levels, and configuring those shifts renders the reflected contributions coherent at an intended receiver, establishing a virtual line-of-sight link without additional radiated power. Under grouped RSMA each active surface is the reflective backbone of one user group, as illustrated in Fig. 1.

Operating such a network requires four coupled decisions at every scheduling interval, namely the assignment of users to surfaces, the phase configuration of each surface, the division of the transmit budget and the apportionment of the group common rate. Together they form a mixed-integer non-convex problem admitting no tractable closed form, for which the established remedy is to decompose the problem and alternate between the resulting subproblems [10], [11]. That remedy is priced per channel realization, the solver retaining nothing between invocations, so that every realization pays the iterative cost anew and a link whose geometry follows user mobility pays it at the rate at which that geometry changes [12].

Reinforcement learning inverts the arrangement. A policy trained once maps an observed state directly onto an allocation, discharging the optimization burden offline and answering each realization with a single forward pass [13], [14]. The

four variables are not independent but form a strict sequential dependency, the assignment fixing the grouping that governs the phase design, the phase design shaping the effective channels that enter the power allocation, and the resulting budget delimiting the feasible common-rate split. That ordering motivates a hierarchical actor-critic decomposition [15], [16], a high-level actor committing to the discrete assignment while dedicated low-level actors resolve the remainder in sequence. The discrete complexity concentrates in the assignment, the number of feasible user-to-surface associations growing exponentially in the user population. A variational quantum circuit (VQC) represents a distribution over that space with a compact set of trainable parameters [17], and because its input would otherwise scale with both the user population and the panel count and so exceed a near-term register [18], an autoencoder [19] learns the latent embedding on which the assignment actually depends, following [20].

Table I places the setting against the prior art. The entries differ in access scheme, panel count, phase resolution and CSI assumption, and the configuration studied here sits at the demanding end of each. That a per-realization solver adapts poorly to a changing environment is itself an established argument [12]. What the table does not record, for the model-based and the learned entries alike, is what producing one allocation costs. That is the question this paper takes up, and it is answered in units that do not depend on the machine the policy runs on.

B. Contributions

The framework developed here addresses that gap from the standpoint of decision cost rather than of achievable rate. It resolves the four variables with a single hierarchical policy that is trained once and thereafter commits to an allocation in one forward pass, and it places a variational quantum circuit at the head of the hierarchy, where the discrete complexity is concentrated. The contributions are summarized as follows.

- We formulate sum-rate maximization for a multi-IRS-assisted grouped RSMA downlink in ISTN, jointly optimizing IRS-user link assignment, phase-shift configuration, transmit power allocation and common-rate distribution as a sequential dependency chain, and balancing aggregate rate against per-user QoS under discrete phase quantization, building blockage and imperfect CSI.
- We propose the HQC-HAC framework, which exploits that dependency. A quantum actor built on a variational quantum circuit resolves the combinatorial IRS-user assignment at the high level, three classical actors resolve the phase, power and common-rate decisions in sequence, and entropy regularization with an annealing schedule sustains exploration across all levels.
- We adapt the AE-VQC pairing of [20] by structuring what the circuit receives. An affinity feature map removes the dependence on the reflecting-element count N , and a dual-branch autoencoder compresses the result into a latent whose two blocks load onto the user and IRS registers indexed by the readout observables, so the compression preserves the user-IRS structure.

- We evaluate two instantiations differing only in the assignment actor and report what a decision costs in units independent of the implementation. The quantum instantiation matches its classical counterpart to within the spread across training seeds while carrying substantially fewer inference parameters, and the invocation count of every learned policy is constant in the user population where that of the solver grows. Ablations attribute this to the compression rather than to the register size, the observable set or the entangling structure, placing the contribution in parameter efficiency and amortized inference rather than in achievable rate.

II. RELATED WORKS

A. IRS-Assisted Grouped RSMA for Sum-Rate Optimization in ISTNs

The foundational framework for IRS-assisted wireless networks was established by Wu and Zhang [21], who demonstrated that jointly optimizing active transmit beamforming at the access point and passive phase shifts at the surface substantially reduces the transmit power required to serve a multiuser MISO downlink.

The broader body of work summarized in Table I leaves the same axes open. Ten of the twelve entries deploy a single reflecting panel, so the reflected coverage cannot follow spatially dispersed user groups, the exceptions being [29], [30]. Eight presume perfect CSI, robust designs being confined to [22], [25], [30], [31], and continuous phase control is near-universal, with quantization appearing only in [25], [26]. Learning-based optimization is rare, adopted in [12], [28], and no entry employs a quantum policy representation. Blockage enters the propagation model in only three cases [22], [23], [29].

Building on that paradigm, Tan et al. [8] extended IRS-assisted transmission to satellite downlinks with a flexible grouped RSMA scheme, partitioning the K users into a reflected group and a direct group so that each group's common rate is bounded by its own weakest member rather than by the globally worst user. Their joint problem over precoding, phase configuration, rate splitting and binary group selection is decomposed into four subproblems and solved iteratively, reporting approximately 29% sum-rate gain over an IRS-assisted RSMA baseline. Three limitations bear on deployment. The design presupposes perfect instantaneous CSI, whereas practical satellite links suffer estimation error from channel aging, Doppler and feedback delay; it admits a single panel and therefore cannot extend reflected coverage across spatially dispersed groups; and its model-based solver must be re-run from scratch at every channel realization, offering no mechanism for adaptation under mobility. HQC-HAC addresses all three.

B. Entropy-Regularized Hierarchical Reinforcement Learning and Hybrid Quantum-Classical Policies

Hierarchical actor-critic (HAC) methods decompose complex decision problems into multiple levels of abstraction, with high-level actors committing to sub-goals that low-level actors

TABLE I: Comparison of existing works on IRS/RIS-assisted communication systems

Ref.	Access	Blockage	CSI	Prop. Path	IRS/RIS			RL	QC	QoS	Obj.
					Scale	Type	Phase				
2019 [21]	MU-BF	–	Perf.	Multi	Single	Passive	Cont.	–	–	Constr.	Min-power
2022 [22]	SDMA	✓	Imperf.	Multi	Single	Passive	Cont.	–	–	–	Sum-rate
2023 [23]	TDMA	✓	Perf.	Multi	Single	Passive	Cont.	–	–	–	WSR
2023 [12]	RSMA	–	Perf.	Multi	Single	STAR	Cont.	✓	–	Constr.	Sum-rate
2025 [24]	H-NOMA	–	Perf.	Single	Single	Passive	Cont.	–	–	Constr.	TA-SR
2025 [25]	RSMA	–	Imperf.	Multi	Single	Passive	Disc.	–	–	Constr.	WSR
2025 [26]	RSMA	–	Perf.	Single	Single	Passive	Disc.	–	–	Constr.	WSR
2025 [27]	MU-BF	–	Perf.	Multi	Single	Passive	Cont.	–	–	–	WSR
2025 [28]	NOMA	–	Perf.	Multi	Single	Passive	Cont.	✓	–	Penalty	ASR
2025 [29]	RSMA	✓	Perf.	Single	Multi	Passive	Cont.	–	–	Constr.	Sum-rate
2026 [30]	RSMA	–	Imperf.	Multi	Multi	Passive	Cont.	–	–	Constr.	WSR
2026 [31]	RSMA	–	Imperf.	Multi	Single	Active	Cont.	–	–	Constr.	Sum-rate
2026 [8]	RSMA	–	Perf.	Single	Single	Passive	Cont.	–	–	Constr.	Sum-rate
Proposed	G-RSMA	✓	Imperf.	Multi	Multi	Passive	Disc.	✓	✓	Penalty	Sum-rate

subsequently fulfill [15], [32]. This temporal and functional decomposition has been demonstrated to accelerate learning in environments with large composite action spaces, including IRS-assisted wireless resource management, where jointly optimizing discrete and continuous variables over extended interaction horizons is otherwise costly to learn directly [16]. Entropy regularization sustains policy diversity during training, the maximum-entropy objective of soft actor-critic [33] penalizing premature collapse to a deterministic policy.

Hybrid quantum-classical reinforcement learning extends this paradigm by replacing or augmenting classical policy networks with variational quantum circuits (VQCs) [17], [34]. VQCs model nonlinear policy mappings through parameterized quantum state transformations with comparatively compact trainable representations, which motivates their study for high-dimensional combinatorial decisions and has prompted direct parameter-based comparisons against neural networks [35]. However, directly encoding high-dimensional wireless system states into a quantum register is impractical given current hardware constraints [18]. Nagy et al. [20] addressed this by jointly training an autoencoder (AE) with a VQC-based policy, where the AE compresses the original observation into a low-dimensional latent space suitable for quantum encoding. This work embeds an AE-VQC actor within an entropy-regularized HAC structure.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network and Association Model

We consider an IRS-assisted grouped rate-splitting multiple access (RSMA) downlink communication system in an integrated satellite-terrestrial network (ISTN). In the considered system, a satellite serves a set of ground users with the assistance of intelligent reflecting surfaces (IRSs), which are deployed on high buildings to enhance the wireless propagation environment and support users experiencing weak or blocked direct satellite-to-user links. The satellite is assumed to act as the central controller, which determines the transmission

strategy, resource allocation, and IRS-assisted communication mode according to the observed channel conditions and user requirements.

Let $\mathcal{K} = \{1, 2, \dots, K\}$ denote the set of downlink ground users, where K is the total number of users. The satellite transmits one independent message to each user, the message intended for user $k \in \mathcal{K}$ being denoted by s_k , so that the scheduled message set is $s = \{s_1, \dots, s_K\}$.

We consider M IRSs deployed on the rooftops of buildings to assist the satellite downlink transmission. Let $\mathcal{M} = \{1, 2, \dots, M\}$ denote the set of IRSs. Each IRS is equipped with N passive reflecting elements, which can independently adjust the phase shift of the incident signal. The IRSs are assumed to be connected to the central controller, and their reflection coefficients are configured according to the channel conditions and user association decisions. For IRS $m \in \mathcal{M}$, the reflection matrix is defined as

$$\Phi_m = \text{diag} (e^{j\theta_{m,1}}, e^{j\theta_{m,2}}, \dots, e^{j\theta_{m,N}}), \quad (1)$$

where $\theta_{m,n}$ denotes the phase shift of the n -th reflecting element at IRS m . Practical surfaces resolve the phase only to a finite number of levels, so with B quantization bits each element selects its shift from the uniformly spaced set [25], [26]

$$\mathcal{Q}_\theta = \left\{ \theta^{(q)} = \frac{2\pi q}{2^B} \mid q = 0, 1, \dots, 2^B - 1 \right\}. \quad (2)$$

To determine the transmission mode of each user, we define an IRS-selection variable for every user $k \in \mathcal{K}$. Since the considered system includes M IRSs, each user can either be served through the direct satellite-to-user link or through one of the IRS-assisted reflected links. Therefore, the IRS-selection variable $\varphi_k \in \{0, 1, \dots, M\}$ is defined for every $k \in \mathcal{K}$ as

$$\varphi_k = \begin{cases} 0, & \text{user } k \text{ served by the direct link,} \\ m, & \text{user } k \text{ served by IRS } m. \end{cases} \quad (3)$$

B. Channel Model

Due to the long propagation distance between the satellite and ground nodes, the satellite-to-user (SU) and satellite-to-IRS (SR) links are dominated by large-scale path loss and rain attenuation following the ITU-R model [36]. The IRS-to-user (RU) link additionally exhibits small-scale Rayleigh fading and distance-dependent path loss [4]. The true channel coefficients of the three links are modeled as

$$\begin{aligned} g_k^{SU} &= \frac{c\sqrt{G_S G_U \epsilon_k^{SU}}}{4\pi f_{SU} d_k^{SU}}, \\ g_m^{SR} &= \frac{c\sqrt{G_S \epsilon_m^{SR}}}{4\pi f_{SR} d_m^{SR}}, \\ g_{m,k}^{RU} &= \sqrt{G_U \cdot \Omega_{sf} \cdot \left(\frac{d_0}{d_{m,k}^{RU}}\right)^\alpha \cdot \epsilon_{m,k}^{RU} \tilde{g}_{m,k}^{sf}}, \end{aligned} \quad (4)$$

for every $m \in \mathcal{M}$ and $k \in \mathcal{K}$, where c is the speed of light; G_S and G_U are the satellite transmit and user receive antenna gains; f_{SU} and f_{SR} are the corresponding carrier frequencies; d_k^{SU} , d_m^{SR} , and $d_{m,k}^{RU}$ denote the satellite-to-user, satellite-to-IRS, and IRS-to-user link distances; ϵ_k^{SU} , ϵ_m^{SR} , and $\epsilon_{m,k}^{RU}$ are the rain-attenuation coefficients modeled as lognormal random variables per the ITU-R standard; Ω_{sf} is the average shadow-fading power gain of the IRS-to-user link; α is the path-loss exponent; d_0 is the reference distance; and $\tilde{g}_{m,k}^{sf} \sim \mathcal{CN}(0, 1)$ represents the small-scale Rayleigh fading component. When the satellite-to-user line of sight is intersected by a building, the affected coefficient g_k^{SU} is scaled by a blocking factor $\beta_{\text{blk}} \in (0, 1]$, so that a blocked user retains a weak direct path rather than losing it entirely. Each channel coefficient is complex, with a random propagation phase uniformly distributed on $[0, 2\pi)$, which is tracked via Doppler compensation at the satellite.

Since the satellite acquires only an imperfect estimate of each channel coefficient, the estimated channel $\hat{g}$ and the true channel g are related by a multiplicative error model:

$$\hat{g} = g + \Delta g, \quad \Delta g = \kappa|g|\varepsilon, \quad \varepsilon \sim \mathcal{CN}(0, 1), \quad (5)$$

where $\kappa \geq 0$ is the relative error coefficient applied uniformly to all three links. This yields the estimated channels $\hat{g}_k^{SU}$, $\hat{g}_m^{SR}$, and $\hat{g}_{m,k}^{RU}$, which are used for all precoding and resource allocation design. The policy is designed on the estimated channels alone, the assignment actor consuming per-user path qualities and the phase actor the estimated cascaded coefficients, whereas the environment scores the achieved rate on the true channels g .

C. Transmission and Decoding

Following the RSMA principle [7], each user message s_k of user $k \in \mathcal{K}_g$ is split into a common sub-message $s_{c,g}$ and a private sub-message $s_{k,p,g}$. However, not all IRSs are necessarily selected by the agent at each decision step. Therefore, the number of active groups is denoted by G , where $G \leq M + 1$. We define the set of scheduled user messages after splitting as

$$\tilde{\mathbf{s}} = [s_{c,1}, s_{c,2}, \dots, s_{c,G}, \{s_{k,p,g}\}_{g \in \mathcal{G}, k \in \mathcal{K}_g}].$$

A single-antenna satellite precludes spatial beamforming, so each stream carries a scalar coefficient, $w_{c,g} \in \mathbb{C}$ for the common stream of group g and $w_{k,p,g} \in \mathbb{C}$ for the private stream of user $k \in \mathcal{K}_g$, and the transmitted signal is

$$x = \sum_{g \in \mathcal{G}} w_{c,g} s_{c,g} + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} w_{k,p,g} s_{k,p,g}, \quad (6)$$

where $s_{c,g}$ is the common stream intended for all users in group $\mathcal{K}_g$, and $s_{k,p,g}$ is the private stream intended only for user k in group g . The scalar precoding coefficients are constrained by the maximum transmit power of the satellite, i.e., $\sum_{g \in \mathcal{G}} |w_{c,g}|^2 + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} |w_{k,p,g}|^2 \leq P_S$, where P_S denotes the maximum transmit power of the satellite.

After receiving y_k , each user decodes its group-specific common stream and then its private stream by successive interference cancellation, following the standard RSMA decoding order [5], [7]. For user $k \in \mathcal{K}_g$, the SINRs for decoding the common stream $s_{c,g}$ and the private stream $s_{k,p,g}$ are respectively given by

$$\begin{aligned} \gamma_{k,c,g} &= \frac{|h_k w_{c,g}|^2}{\sum_{g' \neq g} |h_k w_{c,g'}|^2 + \sum_{g' \in \mathcal{G}} \sum_{j \in \mathcal{K}_{g'}, j \neq k} |h_k w_{j,p,g'}|^2 + \sigma^2}, \\ \gamma_{k,p,g} &= \frac{|h_k w_{k,p,g}|^2}{\sum_{g' \in \mathcal{G}} \sum_{j \in \mathcal{K}_{g'}, j \neq k} |h_k w_{j,p,g'}|^2 + \sum_{g' \neq g} |h_k w_{c,g'}|^2 + \sigma^2}. \end{aligned} \quad (7)$$

Since the common stream $s_{c,g}$ is intended for all users in group $\mathcal{K}_g$, it must be successfully decoded by every user in that group. Therefore, the achievable common rate of group g is limited by the user with the lowest common-stream decoding rate. For user $k \in \mathcal{K}_g$, the common-stream decoding rate and the resulting group common rate are given by

$$R_{k,c,g} = \log_2(1 + \gamma_{k,c,g}), \quad R_{c,g} = \min_{k \in \mathcal{K}_g} R_{k,c,g}. \quad (8)$$

Let $C_{k,g}$ denote the portion of the group common rate allocated to user $k \in \mathcal{K}_g$. The common-rate allocation must satisfy

$$\sum_{k \in \mathcal{K}_g} C_{k,g} = R_{c,g}, \quad C_{k,g} \geq 0. \quad (9)$$

After decoding and removing the common stream, user k decodes its private stream, achieving rate $R_{k,p,g} = \log_2(1 + \gamma_{k,p,g})$. The total achievable rate of user k is thus $R_k = R_{k,p,g} + C_{k,g}$, and the system sum-rate is

$$R_{\text{tot}} = \sum_{k \in \mathcal{K}} R_k = \sum_{g \in \mathcal{G}} R_{c,g} + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} R_{k,p,g}. \quad (10)$$

D. Optimization Problem

The objective is to maximize the system sum-rate by jointly optimizing the IRS-selection variables, the scalar precoding coefficients, the IRS phase-shift matrices, and the common-rate allocation. Specifically, the optimization variables are given by $\varphi = \{\varphi_k\}_{k \in \mathcal{K}}$, $\mathbf{w} = \{w_{c,g}, w_{k,p,g}\}_{g \in \mathcal{G}, k \in \mathcal{K}_g}$,

$\Phi = \{\Phi_m\}_{m \in \mathcal{M}}$, and $\mathbf{C} = \{C_{k,g}\}_{k \in \mathcal{K}_g, g \in \mathcal{G}}$. The sum-rate maximization problem is formulated as

$$(P0): \max_{\varphi, \mathbf{w}, \Phi, \mathbf{C}} R_{\text{tot}} \quad (11a)$$

$$\text{s.t. } \varphi_k \in \{0, 1, \dots, M\}, \quad \forall k \in \mathcal{K}, \quad (11b)$$

$$\sum_{g \in \mathcal{G}} |w_{c,g}|^2 + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} |w_{k,p,g}|^2 \leq P_S, \quad (11c)$$

$$\theta_{m,n} \in \mathcal{Q}_\theta, \quad \forall m \in \mathcal{M}, \forall n \in \mathcal{N}, \quad (11d)$$

$$\Phi_m = \text{diag}(e^{j\theta_{m,1}}, \dots, e^{j\theta_{m,N}}), \quad \forall m, \quad (11e)$$

$$\sum_{k \in \mathcal{K}_g} C_{k,g} = R_{c,g}, \quad \forall g \in \mathcal{G}, \quad (11f)$$

$$C_{k,g} \geq 0, \quad \forall k \in \mathcal{K}_g, \forall g \in \mathcal{G}, \quad (11g)$$

$$\Pr\{R_k \geq R_k^{\min} \mid \hat{g}\} \geq 1 - \epsilon_o, \quad \forall k. \quad (11h)$$

In the above problem (P0), (11b) limits the values of the IRS-selection variables, which determine whether each user is served through the direct satellite-to-user link or one of the available IRS-assisted links. This choice also induces the grouping, so that $\mathcal{K}_g$ and $\mathcal{G}$ in the remaining constraints are themselves functions of φ . (11c) guarantees that the total transmit power of the satellite does not exceed the maximum power budget. (11d) restricts the phase shift of each IRS reflecting element to the discrete quantized phase-angle set, while constraint (11e) defines the corresponding passive IRS phase-shift matrix. Constraint (11f) requires that the allocated common rates fully exhaust the group common rate, i.e., the entire decodable common rate of each group is distributed among its members with no waste. Constraint (11g) ensures non-negativity of the allocated common rates, and constraint (11h) [37], [38] imposes the per-user minimum-rate requirement at an outage level ϵ_o .

IV. PROPOSED HQC-HAC FRAMEWORK

A. Overall Framework

Problem (P0) is a mixed-integer non-convex program [10], [11] whose four variables are not independent. The assignment fixes which users share a group, the grouping conditions the phase design, the resulting effective channels shape the power allocation, and the power budget delimits the feasible common-rate split. HQC-HAC follows that order. At each step the raw channel observation is condensed into the affinity state $\mathbf{s}_t$ of (12), an autoencoder compresses it to a latent $\mathbf{z}_t$, and a variational quantum circuit reading that latent emits the assignment, which is the only genuinely combinatorial variable since $\{0, 1, \dots, M\}^K$ is non-enumerable at the scales considered. Three classical sub-actors then resolve phase, power and common rate in sequence, as shown in Fig. 2. All four actors and a shared critic are trained together by proximal policy optimization [39].

B. Autoencoder-Based Latent State Representation

Encoding the raw observation into a quantum register is impractical, since angle encoding would demand roughly one

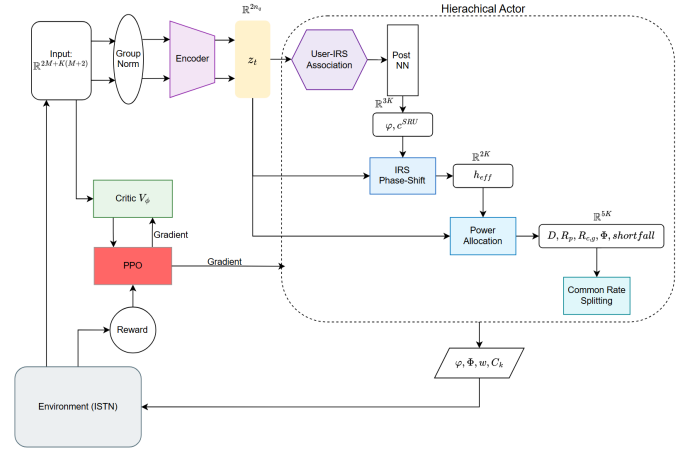


Fig. 2: Overall HQC-HAC hierarchical actor-critic framework. The quantum actor (top level) resolves the discrete IRS-user assignment φ via a VQC, while three sequential classical sub-actors (Phase MLP, Power MLP, and Common-Rate MLP) optimize the remaining continuous variables conditioned on the chosen assignment. A shared classical critic provides state-value estimates for computing advantages across all actors.

qubit per feature, that is 44 qubits at Case 2 and 81 at the $K=15$ scale, placing state-vector simulation with per-step gradients out of reach. The observation is therefore first reduced to an affinity state

$$\mathbf{s}_t = [a_{k,m}, R_k^{\min}, |\hat{g}_k^{SU}|, q_m, p_m], \quad (12)$$

which summarizes each user-IRS pair by a scalar path quality and each IRS by its mean affinity and received power, so that the description no longer grows with the number of reflecting elements. A dual-branch encoder then maps $\mathbf{s}_t$ to $\mathbf{z}_t \in \mathbb{R}^{2n_q}$, one branch per IRS and one shared branch applied to every user, the shared branch making the user block permutation-equivariant in the sense of [40]. The two blocks are loaded onto the IRS and user qubit registers indexed by the readout observables, so the compression preserves the user-IRS structure the assignment depends on rather than its dimension alone, following the hybrid pairing of [20]. A decoder reconstructs the group-normalized state and is trained by

$$\mathcal{L}_{AE} = \frac{1}{2} \|\hat{\mathbf{s}}_t - \bar{\mathbf{s}}_t\|^2, \quad (13)$$

jointly with the rest of the framework after a short reconstruction-only warm-up. The decoder is discarded at inference, so it costs nothing at deployment. The register stays at $n_q=8$ at both scales.

C. Variational Quantum Circuit

The latent vector $\mathbf{z}_t \in \mathbb{R}^{2n_q}$ is processed by a variational quantum circuit whose stages are shown in Fig. 3. Encoding begins with a layer normalization [41] of the latent,

$$\tilde{\mathbf{z}}_t = \gamma \odot \frac{\mathbf{z}_t - \mu_z}{\sigma_z + \epsilon_{LN}} + \mathbf{b}_{LN}, \quad (14)$$

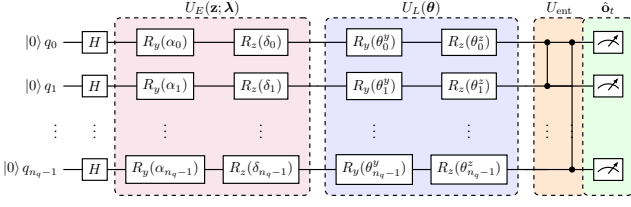


Fig. 3: Variational quantum circuit (VQC) of the quantum actor, shown for representative qubits $q_0, q_1, \dots, q_{n_q-1}$. Following $H^{\otimes n_q}$, each layer applies the encoding block $U_E(\mathbf{z}; \boldsymbol{\lambda})$, variational block $U_L(\boldsymbol{\theta})$, and entangling block U_{ent} , with data re-uploading across layers. The measurement block marks the readout stage, whose observable set is defined in (17).

whose learnable affine restores the DC component a plain zero-mean normalization removes. The normalized entries are mapped to per-qubit rotation angles through trainable scales $\boldsymbol{\lambda}_y, \boldsymbol{\lambda}_z \in \mathbb{R}^{n_q}$,

$$\alpha_i = \pi \tanh(\lambda_i^y \tilde{z}_{2i}), \quad \delta_i = \pi \tanh(\lambda_i^z \tilde{z}_{2i+1}), \quad (15)$$

for $i = 0, \dots, n_q - 1$, so that each latent pair drives one qubit. A Hadamard layer prepares a uniform superposition, after which L identical layers apply the encoding block U_E , a parametrized rotation layer $U_L(\boldsymbol{\theta}_\ell)$ and a fixed entangling block U_{ent} ,

$$|\psi_t\rangle = \left[\prod_{\ell=L}^1 U_{\text{ent}} U_L(\boldsymbol{\theta}_\ell) U_E(\tilde{\mathbf{z}}_t; \boldsymbol{\lambda}) \right] H^{\otimes n_q} |0\rangle^{\otimes n_q}. \quad (16)$$

Re-applying U_E at every layer is data re-uploading [42], which lifts the frequency content the circuit can represent [43]; the entangling block carries no parameters, so training acts on $\boldsymbol{\lambda}$ and $\boldsymbol{\theta}$ alone. Rather than reading the full state $|\psi_t\rangle$ of (16), the circuit is measured through a structured set of at most two-qubit Pauli observables,

$$\hat{\mathbf{o}}_t = [\langle Z_u \rangle, \langle Z_u Z_r \rangle, \frac{1}{|\mathcal{U}|-1} \sum_{u' \neq u} \langle Z_u Z_{u'} \rangle, \langle Z_r \rangle] \in [-1, 1]^{N_Q}, \quad (17)$$

with $N_Q = (n_q - M)(M + 2) + M$. Low-weight observables keep the readout estimable from a practical shot budget and avoid the exponentially flat cost landscapes that global measurements induce [44], [45]. A circuit alone does not define a policy, so a linear layer maps the readout, concatenated with the latent as $\mathbf{h}_t = [\mathbf{z}_t || \hat{\mathbf{o}}_t]$, onto one categorical distribution per user [17],

$$\boldsymbol{\pi}_k = \text{softmax}\left(\tau [\mathbf{W} \mathbf{h}_t + \mathbf{b}]_{k(M+1):(k+1)(M+1)}\right), \quad (18)$$

where $\mathbf{W}$ and $\mathbf{b}$ are trainable, τ is a trainable inverse temperature, and the skip term $\mathbf{z}_t$ restores pre-circuit information the N_Q -entry readout does not carry. The K blocks are softmaxed independently, so an assignment is drawn as $\varphi_k \sim \boldsymbol{\pi}_k$ for every user.

D. Classical Sub-Actors

Three classical actors consume the assignment and resolve the remaining variables. The *Phase MLP* selects a quantization

Algorithm 1 HQC-HAC training

Require: $\Theta = \{\boldsymbol{\omega}, \boldsymbol{\eta}, \xi, \Theta_\Phi, \Theta_{\mathbf{w}}, \Theta_{\mathbf{C}}\}$, ψ , $E=12$, $T=200$, $N_{\text{ep}}=6$

- 1: $\boldsymbol{\omega} \leftarrow \arg \min \mathcal{L}_{AE}$ $\triangleright \Theta \setminus \boldsymbol{\omega}, \psi$ frozen
- 2: $\boldsymbol{\eta}, \xi \leftarrow \arg \min \mathbb{E}[-\log \pi_\varphi(\boldsymbol{\varphi}^*)]$
- 3: $\Theta_\Phi \leftarrow \arg \min \mathbb{E}[-\log \pi_\Phi(\Phi^* | \boldsymbol{\varphi}^*)]$
- 4: **repeat**
- 5: $\mathcal{B} \leftarrow \emptyset$
- 6: **for** $e = 1 : E$ **do**
- 7: **for** $t = 1 : T$ **do**
- 8: $\mathbf{s}_t \leftarrow \mathbf{s}(\hat{g}_t)$, $\mathbf{z}_t \leftarrow \text{AE}_\omega(\mathbf{s}_t)$
- 9: $\boldsymbol{\varphi}_t \sim \pi_\varphi(\cdot | \mathbf{s}_t)$
- 10: $\Phi_t \sim \pi_\Phi(\cdot | \hat{g}_t, \mathbf{z}_t, \boldsymbol{\varphi}_t)$
- 11: $\mathbf{w}_t \sim \pi_{\mathbf{w}}(\cdot | \hat{g}_t, \mathbf{z}_t, \boldsymbol{\varphi}_t, \Phi_t)$
- 12: $\mathbf{C}_t \sim \pi_{\mathbf{C}}(\cdot | \boldsymbol{\varphi}_t, \Phi_t, \mathbf{w}_t)$
- 13: $r_t \leftarrow r(\mathbf{a}_t; g_t)$, $\mathbf{s}_{t+1} \sim \mathcal{P}(\cdot | \mathbf{s}_t, \mathbf{a}_t)$
- 14: $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r_t, V_t, \{\log \pi_j^{\text{old}}\}_j)\}$
- 15: **end for**
- 16: **end for**
- 17: $\hat{A} \leftarrow \text{GAE}(\mathcal{B})$, $\hat{A} \leftarrow (\hat{A} - \mu_{\hat{A}}) / \sigma_{\hat{A}}$
- 18: **for** $n = 1 : N_{\text{ep}}$ **do**
- 19: **for all** $\mathcal{B}_m \subset \mathcal{B}$ **do**
- 20: $\hat{\mathbf{J}} \leftarrow \text{SPSA}(\partial \hat{\mathbf{o}}_t / \partial \boldsymbol{\eta})$
- 21: $\nabla_{\boldsymbol{\eta}} \mathcal{J}_\varphi \leftarrow \hat{\mathbf{J}}^\top \nabla_{\hat{\mathbf{o}}_t} \mathcal{J}_\varphi$
- 22: $\Theta \leftarrow \Theta + \alpha \nabla_{\Theta} \mathcal{J}$
- 23: $\psi \leftarrow \psi - \alpha_\psi \nabla_\psi \mathcal{L}_V$
- 24: **end for**
- 25: **end for**
- 26: $\beta \leftarrow \max(\beta_{\min}, \beta - \Delta\beta)$, $\alpha \leftarrow \text{decay}(\alpha)$, $(\mu, \sigma) \leftarrow \text{PopArt}(G)$
- 27: **until** episode budget exhausted

level for every reflecting element from the estimated cascaded coefficients and the latent, sharing one weight set across elements so that it does not depend on N . The *Power MLP* factors the budget through three Dirichlet heads, a two-way split between the common and private pools followed by an allocation within each, so the executed action is the quantity the policy gradient scores. The *Common-Rate MLP* reads the demands $R_k^{\min}$, the private rates already achieved $R_{k,p,g}$, the group common rate $R_{c,g}$, the shortfalls ς_k and the assignment φ_k , and draws a Dirichlet vector apportioning the common rate within each group, so $\sum_{k \in g} C_k \leq R_{c,g}$ holds by construction. All three share the advantage of Section IV-E and are updated by the same clipped surrogate.

E. Entropy-Regularized Hierarchical Actor-Critic Training

The task is cast as a Markov decision process $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma_d)$ whose state is the affinity vector of (12) and whose action is $\mathbf{a}_t = (\varphi, \Phi, \mathbf{w}, \mathbf{C})$. The hierarchy realizes that action as a chain,

$$\begin{aligned} \pi(\mathbf{a}_t | \mathbf{s}_t, \hat{g}) &= \pi_\varphi(\varphi | \mathbf{s}_t) \pi_\Phi(\Phi | \hat{g}, \mathbf{z}_t, \varphi) \\ &\quad \times \pi_{\mathbf{w}}(\mathbf{w} | \hat{g}, \mathbf{z}_t, \varphi, \Phi) \pi_{\mathbf{C}}(\mathbf{C} | \varphi, \Phi, \mathbf{w}), \end{aligned} \quad (19)$$

each factor conditioned on the decision-time context and on the decisions taken upstream, which gives $\log \pi = \sum_j \log \pi_j$ so the surrogate below applies per actor. The process has no absorbing state, an episode being a fixed-length truncation. The reward links the objective (11a) to the QoS

constraint (11h) through a soft quadratic penalty,

$$r_t = R_{\text{tot}} - \lambda_D \sum_{k \in \mathcal{K}} \left(\frac{\max(0, R_k^{\min} - R_k)}{R_k^{\min} + \epsilon_q} \right)^2, \quad (20)$$

which vanishes when every user meets its demand and otherwise grows with the normalized shortfall, steering the policy toward the constrained optimum of (P0) without a hard feasibility boundary. The rates entering (20) are those achieved on the true channel g , whereas every factor of (19) is conditioned on the estimate $\hat{g}$ alone. Each actor is updated with the clipped PPO surrogate [39] combined with an entropy bonus and, for the quantum actor, the reconstruction loss (13),

$$\begin{aligned} \mathcal{J}_\varphi &= \mathcal{L}_{\text{PPO}}^\varphi + \beta H(\pi_\varphi) - w_\varphi \mathcal{L}_{\text{sup}}^\varphi - \alpha_{AE} \mathcal{L}_{AE}, \\ \mathcal{J}_j &= \mathcal{L}_{\text{PPO}}^j + \beta H(\pi_j) - w_j \mathcal{L}_{\text{sup}}^j, \quad j \in \{\Phi, \mathbf{C}\}, \quad (21) \\ \mathcal{J}_w &= \mathcal{L}_{\text{PPO}}^w + \beta H(\pi_w), \end{aligned}$$

where $\mathcal{L}_{\text{sup}}^j = -\log \pi_j(a_j^*)$ is a supervised oracle teacher evaluated at a reference action and weighted by w_j , the power actor carrying none. A single critic $V(\mathbf{s}_t; \psi)$ supplies the advantage $\hat{A}_t$ common to all four actors by generalized advantage estimation [46], stabilized by PopArt value-target normalization [47]. The quantum parameters $\boldsymbol{\eta}$ enter the policy only through the circuit, and their Jacobian $\partial \hat{a}_t / \partial \boldsymbol{\eta}$ is estimated by simultaneous perturbation stochastic approximation [48] at a cost of $2R$ circuit evaluations per sample irrespective of $\dim(\boldsymbol{\eta})$, against $2 \dim(\boldsymbol{\eta})$ for the parameter-shift rule. Every other gradient is exact. The entropy coefficient and the learning rates are annealed on fixed schedules, and joint training is preceded by a supervised warm-up of the assignment and phase actors on the same oracle references. Algorithm 1 summarizes the procedure.

V. EXPERIMENT

A. Simulation Setup and Evaluation Protocol

The framework is evaluated at two network scales, $(K, M) = (5, 1)$ and $(10, 2)$, under the channel model of Section III. Table II lists the simulation and training parameters. The two scales share every value in it except the user and panel counts and the circuit depth, $L = 2$ and $L = 3$ respectively.

Four baselines are used. AO re-solves every state by alternating optimization over the same four variables and is reported at the cheapest setting reaching its objective plateau. Greedy routes each user to the panel with the strongest estimated path and divides power evenly; Random draws every variable uniformly. PPO-flat ports the flat policy of [12], one network emitting all four variables from the unaggregated channel under a single PPO ratio. Only that architectural choice is taken from the original. It is given GAE advantages where [12] uses a plain Monte-Carlo return, an annealed entropy bonus it does not have, KL early stopping without which most samples clip and the gradient vanishes, the same critic and PopArt normalisation as the hierarchical actors, and a state augmented with the per-user demand and blockage indicators their system has no counterpart for. Its learning rate was re-tuned from their 10^{-4} to $3 \cdot 10^{-4}$ and swept again

at $(10, 2)$, where it fails at both, and at 305K and 725K parameters it is the widest network of any arm.

Every learned agent performance is a mean over five training seeds. The checkpoint of a run is the one maximizing episode return subject to a 90% QoS floor, selected on three validation environments and re-scored on three disjoint test environments with actions taken greedily. The floor keeps the methods comparable, since PPO-flat trades service for rate monotonically through training whereas the hierarchical actors do not, so an unconstrained argmax would place each method at a different kind of operating point. At the extreme power budgets of Table VI no checkpoint of any arm clears the floor, and the selection falls back there to unconstrained maximum return.

B. Sum-Rate and QoS Performance

Fig. 4 shows the training trajectories and Table III the corresponding greedy-inference results. Rows are compared on reward rather than on R_{tot} , since the policies settle at different QoS levels and a sum-rate obtained by underserving users is not comparable with one that meets the demand.

Against the per-state solver, which attains 100% QoS by construction, both learned actors gain about 31% in reward at $(5, 1)$ and 2.2% at $(10, 2)$, and both clear the strongest heuristic by roughly 59% and 20%. The margin over AO therefore narrows sharply with scale. At $(10, 2)$ the quantum actor's 337.3 ± 8.8 lies within one standard deviation of the solver's 330.1, so only the classical instantiation separates from AO there with confidence.

The two instantiations perform closely at both scales, their reward gap staying well inside the spread across training seeds. This is what the prior for parametrized quantum policies anticipates, namely that they match rather than exceed classical ones in task performance [17], [49]. Where they differ is in how service is distributed. At $(10, 2)$ the quantum actor holds $94.4 \pm 1.4\%$ QoS against $92.1 \pm 2.0\%$, the only separation in this study exceeding two standard errors.

PPO-flat is the uneven one, competitive at $(5, 1)$ where it reaches 481.2 ± 6.4 at 96.0% QoS, the highest of any learned policy, yet falling at $(10, 2)$ to 0.557 bps/Hz and 47.2%. Section V-E identifies the mechanism behind that collapse.

C. Deployment Cost

Table IV decomposes both actors by role. The comparison is meaningful at the level of the whole assignment actor, since the circuit reaches its logits through a softmax layer that the classical policy network does not require, so the 80 and 96 parameters of the quantum core are not by themselves a substitute for the 1,410 and 2,230 of that network. Counted over the whole actor at inference, where the encoder is included and the reconstruction decoder discarded, the quantum pipeline is $9.2\times$ and $6.0\times$ smaller, and the complete policy is $18.4\times$ and $33.2\times$ smaller than the flat baseline.

Table V reports what a decision and a training episode cost. At inference the solver spends 23.93 and 53.74 ms per decision against between 0.42 and 2.78 ms for the learned policies, but the more durable statement is the invocation count. AO

TABLE II: Simulation and Training Hyperparameters

Parameter	Value	Parameter	Value
<i>System, channel, and quantum actor</i>		<i>Reinforcement learning and training</i>	
Users K / IRS M (Case 1/2)	(5, 1) / (10, 2)	Discount factor γ_d	0.95
Elements per IRS N	24	GAE parameter λ_{GAE}	0.9
Satellite power P_S	50 dBm	PPO clip ϵ_c / epochs	0.2 / 6
Carrier frequency f / altitude h_{SR}	1.58 GHz / 800 km	Rollout episodes / mini-batch	12 / 64
Antenna gains G_S, G_U	60 dBi	Entropy $\beta_0 \rightarrow \beta_{\min}$	$10^{-3} \rightarrow 3 \times 10^{-4}$
Noise n_0 (mean, variance)	(−30, 10) dBW	QoS penalty λ_D / guard ϵ_q	$1.5 / 10^{-3}$
Rain attenuation (μ, σ^2)	(4 dB, 0.1)	Reward noise averaging	16
CSI error coefficient κ	0.05	LR (AE ω / VQC η / layer ξ)	$10^{-4} / 10^{-4} / 3 \times 10^{-4}$
Phase quantization bits B	2 (4 levels)	LR (phase, power / C_k / critic)	$10^{-4} / 5 \times 10^{-5} / 3 \times 10^{-4}$
QoS demand $R_k^{\min}$	0.05 bps/Hz	Episodes / steps per episode	20,000 / 200
LoS radius R_{LoS} / IRS height	0.5 km / 20 m	Optimizer	Adam
Blocking factor β_{blk}	0.1	Gradient-norm clip (AE / actors)	0.1 / 0.5
User speed	1.5 m/s		
Qubits n_q / latent dim $2n_q$	8 / 16 (DNN latent 24)		
Variational layers L (Case 1/2)	2 / 3		
Data re-uploading	enabled		
SPSA step c_s / repetitions	0.1 / 16		
Measurement shots S (train / eval)	1500 / 3000		
Autoencoder loss weight α_{AE}	0.5		

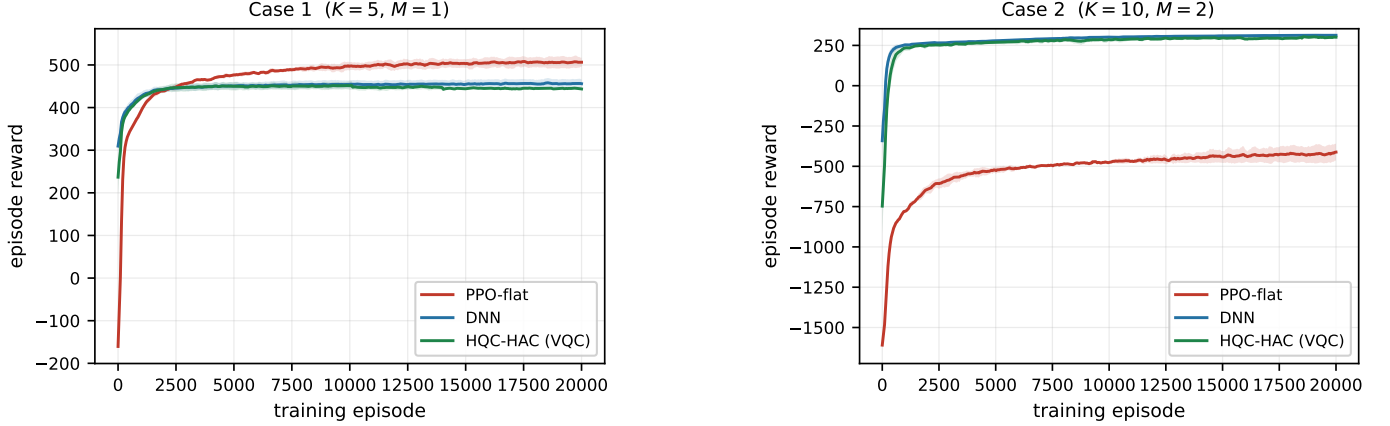


Fig. 4: Training curves for both problem sizes, showing mean episode return over five training seeds under stochastic actions, with shaded regions indicating one standard deviation. Resumed training runs are stitched across segments; for $(K, M) = (5, 1)$, the VQC seed budget decreases after 10,000 episodes as completed seeds are excluded from the mean, while PPO-flat at $(10, 2)$ averages two runs. The curves are not directly comparable with Table III, which reports greedy inference at checkpoints selected under a 90% QoS floor.

calls the rate kernel 482 and 896 times per action, growing with the user population, whereas every learned policy calls it three times irrespective of scale. That count is independent of language, machine, and, for the quantum actor, of the fact that its circuit is simulated rather than executed.

The quantum actor requires 18.4 and 25.2 s per episode against 1.7 and 2.7 for its classical counterpart and under 1.2 for the flat baseline, and issues about $3.9 \cdot 10^4$ circuit evaluations per episode. The solver requires no training whatsoever, so its entire cost falls at inference. That asymmetry is precisely the trade an amortized policy makes, and it is favourable only where the deployed system decides far more often than it is retrained.

D. Robustness to Distribution Shift

Table VI evaluates every policy under thirteen shifts applied after training. Quadrupling the CSI error from $\kappa = 0.05$ to 0.20 costs the quantum actor 0.04% of its sum-rate at $(5, 1)$ and 0.7% at $(10, 2)$, and raising the noise variance from 10 to 16 dB² costs 1.1% and 2.3%. Both learned actors therefore degrade gracefully under exactly the perturbations that imperfect CSI is meant to model.

Transmit power is a different axis, a deployment parameter rather than a runtime one, so those rows are retrained at their own budget. Both learned actors retain an advantage down to 50 dBm, while at 40 dBm and under combined-A the solver leads on both metrics at once.

PPO-flat fails discontinuously. At $(10, 2)$ it falls below the

TABLE III: Performance under greedy inference at $P_S = 50$ dBm. All policies are evaluated on the same 600 test states (3 environment seeds $\times$ 200 steps). Checkpoints are selected on disjoint validation environments by maximizing greedy episode reward subject to a 90% QoS requirement. Learned rows are mean $\pm$ standard deviation over five independent training seeds, the solvers being deterministic.

Policy	Reward	R_{tot}	QoS	Shortfall
<i>Case 1 ($K=5, M=1$)</i>				
HQC-HAC (VQC)	490.3 ± 22.1	2.482 ± 0.100	$92.4 \pm 3.2\%$	$1.29\%/6.5 \cdot 10^{-4}$
DNN	494.8 ± 24.3	2.501 ± 0.121	$94.6 \pm 2.7\%$	$1.00\%/5.0 \cdot 10^{-4}$
PPO-flat	481.2 ± 6.4	2.417 ± 0.035	$96.0 \pm 3.8\%$	$0.23\%/1.1 \cdot 10^{-4}$
AO	375.1	1.876	100.0%	0.00%/0
Greedy	309.1	1.549	99.8%	$0.06\%/0.3 \cdot 10^{-4}$
Random	-307.6	0.579	47.8%	$33.81\%/169 \cdot 10^{-4}$
<i>Case 2 ($K=10, M=2$)</i>				
HQC-HAC (VQC)	337.3 ± 8.8	1.742 ± 0.021	$94.4 \pm 1.4\%$	$0.88\%/4.4 \cdot 10^{-4}$
DNN	337.5 ± 2.3	1.747 ± 0.011	$92.1 \pm 2.0\%$	$1.08\%/5.4 \cdot 10^{-4}$
PPO-flat [†]	-367.1 ± 53.7	0.557 ± 0.039	$47.2 \pm 3.1\%$	$23.42\%/117 \cdot 10^{-4}$
AO	330.1	1.651	100.0%	0.00%/0
Greedy	281.2	1.434	99.4%	$0.21\%/1.1 \cdot 10^{-4}$
Random	-1582.9	0.315	19.6%	$63.90\%/320 \cdot 10^{-4}$

[†]Two runs at learning rates $3 \cdot 10^{-4}$ and $1 \cdot 10^{-4}$ rather than a five-seed mean.

TABLE IV: Parameter budget across problem sizes (K, M). The upper block reports the *assignment actor*, where the VQC and DNN variants are directly substitutable; training parameters additionally include the autoencoder decoder, which is discarded after training. The lower block reports the complete inference policy, the level at which PPO-flat is comparable because it jointly outputs all four decision variables.

	(5, 1)	(10, 2)
<i>Assignment actor</i>		
VQC core	80	96
Softmax policy layer	391	1,291
AE encoder	272	446
AE decoder (<i>training only</i>)	561	2,268
VQC, inference	743	1,833
VQC, training	1,304	4,101
DNN	6,810	11,086
DNN/VQC (inference / training)	$9.2 \times / 5.2 \times$	$6.0 \times / 2.7 \times$
<i>Complete policy, inference</i>		
HQC-HAC (VQC)	16,538	21,831
Classical (DNN)	23,629	32,108
PPO-flat	305,016	725,239
PPO-flat/HQC-HAC	$18.4 \times$	$33.2 \times$

solver at ten of thirteen shifts, and its column is static across the base point and all five CSI and noise shifts, so the failure is one break at the base operating point rather than a response to the shifts. The same table nevertheless places it ahead of every other method once the budget is generous, at 503.8 and 2036.7 against 438.7 and 454.3 for the classical actor at 60 and 70 dBm. The hierarchical actors fall below the solver only where the solver itself leaves 100% QoS. Daggers [†] mark operating points below the 90% floor, twenty-two of the seventy-eight learned entries. At (10, 2) the quantum actor

TABLE V: Cost of one decision at inference and one training episode. The upper half reports per-decision inference cost, while the lower half reports per-episode training cost. Within each half, the first block gives wall-clock time and the second gives implementation-independent invocation counts. Wall-clock values are medians. AO is re-solved independently for each state and is not trained; hence, training entries do not apply.

		AO	PPO-flat	DNN	VQC
<i>Inference, per decision</i>					
Latency (ms)	(5, 1)	23.93	0.42	0.60	1.94
	(10, 2)	53.74	0.66	0.85	2.78
Rate-kernel evals	(5, 1)	482	3	3	3
	(10, 2)	896	3	3	3
<i>Training, per episode</i>					
Wall-clock (s)	(5, 1)	n/a	0.87	1.72	18.38
	(10, 2)	n/a	1.19	2.72	25.23
Circuit evals	(5, 1)	n/a	n/a	n/a	$3.9 \cdot 10^4$
	(10, 2)	n/a	n/a	n/a	$3.9 \cdot 10^4$

clears it at all thirteen shifts and PPO-flat at one.

E. Budget Allocation Policy

The comparisons so far report what each policy achieves; Fig. 5 reports how, as the Lorenz curve of the private power across users at four transmit budgets. The solver lies on the diagonal at every rung, Gini 0.00 throughout, because it divides private power equally by construction. Whatever the learned policies gain on the power axis of Section VI is gained here.

For the hierarchical actors the dependence on the budget is not monotone. Concentration attains a minimum at intermediate budgets, 0.10 and 0.07 at (5, 1) and 60 dBm, and rises at both extremes, to 0.50 for either actor at 40 dBm and to 0.14 and 0.19 at 70 dBm. At the low end no even split brings any user to its demand, so serving a subset well is what the reward prefers; at the high end the shortfall penalty is capped at λ_D per withheld user while the rate term grows with P_S , so withholding eventually pays for itself. This is the trade-off of Section VI seen along the power budget rather than along λ_D .

PPO-flat shows only the second rise, and without bound. At 40 dBm it is the least concentrated of the three learned policies at both scales; at 70 dBm it places the entire private budget on one user. It answers the incentive that rewards concentration where the budget is large enough that the demand is met whatever the allocation, and not the one that punishes it where the budget is tight. That single mechanism accounts for both of its anomalies, the collapse at (10, 2) in Table III and the lead at 60 and 70 dBm in Table VI, each being the same policy answered by a different budget.

VI. RATE-QOS FRONTIER

The policies of Section V-B settle at different service levels, so the reference they are measured against has to be defined

TABLE VI: Zero-shot generalization under distribution shifts for both cases. Policies are trained at the base operating point ($P_S = 50$ dBm, $\kappa = 0.05$, $\sigma^2 = 10$ dBW, user speed 1.5 m/s) and evaluated without retraining under shifted environments, while AO is re-optimized for each shifted state. Combined-A denotes $\{P_S = 40$ dBm, $\kappa = 0.10$, speed = 3 m/s} and Combined-B denotes $\{P_S = 60$ dBm, $\kappa = 0.20$, speed = 6 m/s, $\sigma^2 = 16$ dBW}. Operating points with $P_S \neq 50$ dBm are retrained and reported separately, as transmit power is a deployment parameter rather than a runtime environmental variable. The N -shift rows are zero-shot for the hierarchical actors, whose affinity map and phase sub-actor do not depend explicitly on N ; PPO-flat is retrained because its unaggregated channel input dimension changes with N . The 90% QoS floor is applied only to base-row checkpoint selection; shifted rows use unconstrained argmax return. A dagger indicates QoS below 90%, for which the corresponding rate is not directly comparable with entries satisfying the QoS floor.

Shift	AO			PPO-flat			DNN			HQC-HAC (VQC)		
	R_{tot}	QoS	Reward	R_{tot}	QoS	Reward	R_{tot}	QoS	Reward	R_{tot}	QoS	Reward
<i>Case 1</i> ($K = 5, M = 1$)												
Base	1.8366	100.0%	367.3	2.4658	93.6%	489.3	2.4829	94.4%	491.1	2.4653	92.3%	486.5
$P_S = 40$	1.3278	100.0%	265.3	1.0812	92.7%	197.0	1.8350	76.9% [†]	329.1	1.7917	78.8% [†]	321.5
$P_S = 60$	1.9928	100.0%	398.6	2.9638	97.5%	587.3	2.8024	94.3%	556.5	2.7942	95.3%	556.3
$P_S = 70$	2.0809	100.0%	416.2	10.2956	99.9%	2059.0	3.8151	81.7% [†]	596.2	3.9302	80.5% [†]	601.8
$N = 16$	1.8216	100.0%	364.3	2.1698	92.5%	430.0	2.3710	93.2%	466.3	2.3449	92.1%	461.9
$N = 32$	1.9105	100.0%	382.1	2.6684	90.4%	531.5	2.5764	96.0%	512.2	2.5720	92.9%	508.8
$\kappa = 0.10$	1.8369	100.0%	367.4	2.4667	93.6%	489.5	2.4808	94.3%	490.2	2.4661	92.3%	486.7
$\kappa = 0.15$	1.8365	100.0%	367.3	2.4662	93.5%	489.4	2.4722	94.1%	487.1	2.4665	92.2%	486.5
$\kappa = 0.20$	1.8366	100.0%	367.3	2.4659	93.5%	489.2	2.4588	93.0%	478.7	2.4654	92.2%	486.3
$\sigma^2 = 13$	1.8232	100.0%	364.6	2.4425	92.9%	484.1	2.4664	94.0%	487.4	2.4507	92.0%	483.0
$\sigma^2 = 16$	1.8104	100.0%	362.1	2.4203	92.1%	478.8	2.4506	93.7%	483.6	2.4337	91.7%	479.0
Combined-A	1.1427	98.9%	225.2	1.0048	91.2%	173.2	1.7265	78.0% [†]	308.8	1.7431	76.7% [†]	307.0
Combined-B	1.7818	100.0%	356.3	2.5592	97.4%	504.7	2.8937	95.4%	574.6	2.8356	95.4%	564.5
<i>Case 2</i> ($K = 10, M = 2$)												
Base	1.6515	100.0%	330.3	0.5874	54.4% [†]	-326.5	1.7522	91.5%	337.4	1.7397	94.4%	337.1
$P_S = 40$	1.2979	93.8%	240.1	0.8579	76.3% [†]	-244.1	0.9825	92.7%	162.0	0.9786	93.2%	162.3
$P_S = 60$	1.7418	100.0%	348.4	2.6097	89.7% [†]	503.8	2.2359	87.4% [†]	438.7	2.1907	92.0%	432.9
$P_S = 70$	1.7581	100.0%	351.6	10.1880	98.0%	2036.7	2.2988	87.7% [†]	454.3	2.2860	91.1%	453.3
$N = 16$	1.6341	100.0%	326.8	0.5771	51.5% [†]	-283.0	1.6688	89.8% [†]	309.5	1.6594	92.6%	308.6
$N = 32$	1.6575	100.0%	331.5	0.5138	43.5% [†]	-397.4	1.7754	93.1%	345.4	1.7568	95.3%	345.1
$\kappa = 0.10$	1.6513	100.0%	330.3	0.5874	54.4% [†]	-326.4	1.7516	91.5%	337.4	1.7380	94.3%	336.3
$\kappa = 0.15$	1.6513	100.0%	330.3	0.5873	54.4% [†]	-326.5	1.7495	91.5%	336.2	1.7321	94.2%	334.5
$\kappa = 0.20$	1.6508	100.0%	330.2	0.5872	54.4% [†]	-326.4	1.7478	91.4%	335.5	1.7239	94.0%	331.9
$\sigma^2 = 13$	1.6410	100.0%	328.2	0.5909	55.3% [†]	-337.4	1.7321	91.2%	332.3	1.7157	94.2%	331.4
$\sigma^2 = 16$	1.6309	100.0%	326.2	0.5943	56.0% [†]	-347.2	1.7134	90.8%	327.3	1.6942	93.9%	326.3
Combined-A	1.0949	93.6%	201.5	0.8163	72.8% [†]	-317.3	0.9646	91.7%	159.2	0.9632	92.3%	158.4
Combined-B	1.6233	100.0%	324.4	2.7734	84.2% [†]	503.3	2.3042	89.8% [†]	453.0	2.2558	93.5%	445.1

at a service level too. For a fixed environment let

$$F(q) = \max_{\varphi, \Phi, \mathbf{w}, \mathbf{C}} \mathbb{E}[R_{\text{tot}}] \quad \text{s.t.} \quad \mathbb{E}[\text{QoS}] \geq q, \quad (22)$$

the largest mean sum-rate attainable at mean QoS q under the constraints of (P0). The constraint binds the mean, as Table III reports it. Constraining every state separately defines a different and strictly lower quantity, since it forbids trading service between states, and evaluating one against the other is a common way to overstate how close a policy sits. We obtain F on the 600 test states of Table III by enumerating every assignment, solving the phase exactly for each, sweeping the power split on a grid, filling C_k by demand, and tracing the upper envelope by Lagrangian sweep, with decisions taken on $\hat{g}$ and scored on g as in training. Because the power search is a grid rather than a full optimization, F is an achievable lower bound on the true frontier.

Fig. 6 shows the frontier is shallow across the high-service region and steepens only once QoS is allowed to fall, the signature of an interference-limited link, in which concentrating

power raises one user's rate and everyone else's interference with it. Serving every user is not merely costly but sometimes infeasible, attainable in 99.5% of states at (5, 1) and in 87.3% at (10, 2). Of the 3.38 and 2.15 bps/Hz the frontier reaches at full service, the equal-split identity accounts for 1.61 and 1.52; the remainder is bought by dividing power unequally, the axis on which a solver that splits private power evenly by construction cannot compete.

All four policies lie well below the frontier, and their distances from it are not read the same way. The solver is pinned to the right-hand edge, because its common-rate allocation is ordered by service and it therefore ends at full QoS whatever the channel, and that edge is where the frontier is lowest. Proximity to F is accordingly not the figure of merit, since F maximizes rate at a fixed service level whereas every policy here maximizes the QoS-penalized reward of (20). The two targets are distinct, and the point the reward itself selects lies at 98.1% and 95.0% QoS rather than at 100%. Measured on that objective the learned actors reach 73.1% and 79.4% of

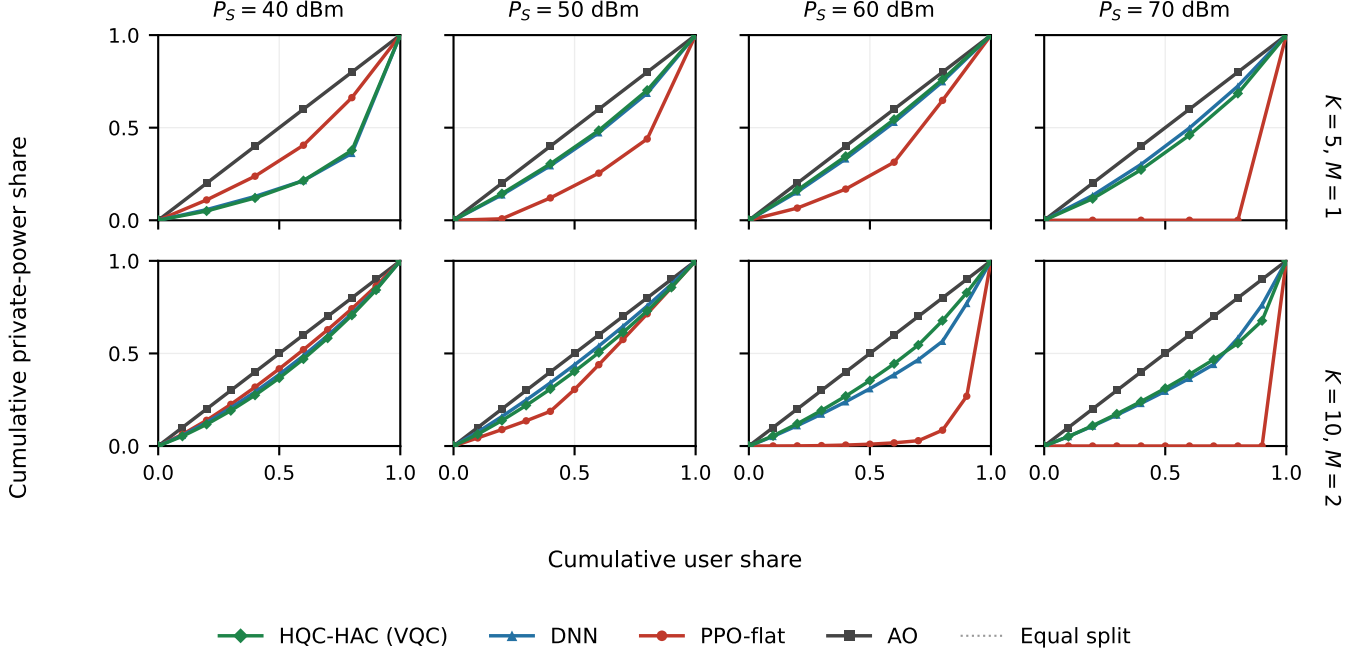


Fig. 5: Lorenz curves of private-power allocation across transmit-power budgets. Users are ordered by received private power; the diagonal denotes equal allocation, and the area between a curve and the diagonal gives the Gini coefficient. The common stream is omitted because it is allocated per group rather than per user. Curves use the greedy checkpoints of Table VI.

the attainable return, against 55.4% and 77.7% for the solver.

The operating point is therefore chosen by λ_D rather than arrived at by accident. Withholding service from one user costs $\lambda_D \hat{p}$ with $\hat{p} = (R_k^{\min} / (R_k^{\min} + \epsilon_q))^2$, so the serve-all optimum takes over above

$$\lambda_{\text{crit}} = \frac{V_{\text{rate}}^* - R^*(K)}{(K-1)\hat{p}}, \quad V_{\text{rate}}^* = \mathbb{E}[\log_2(1 + \Gamma_{\max})], \quad (23)$$

V_{rate}^* being the rate of placing the whole budget on the strongest link, the one configuration free of intra-system interference. Measured, $\lambda_{\text{crit}} = 1.74$ at (5, 1) and 1.01 at (10, 2), which places the training value $\lambda_D = 1.5$ on opposite sides of the two cases. At the smaller scale the reward's global optimum is the degenerate single-user concentration, and entropy regularization is what holds the policy in the serve-all region; at the larger scale the reward already prefers serving everyone.

VII. DISCUSSION

The evidence assembled here supports a claim of feasibility and efficiency rather than of quantum advantage on the task itself. At both scales the variational actor and the classical actor it replaces are separated by less than a quarter of the spread across training seeds, which is what the prevailing prior for parametrized quantum policies anticipates [17], [49]. What distinguishes them is the budget at which that performance is obtained. Set against a solver re-run at every channel realization [8], the learned policy converts a recurring per-state cost into a single offline one.

The measurements also separate what the design owes to compression from what it owes to the circuit. The autoencoder

is indispensable because it decouples the register from the problem size, a direct angle encoding of the affinity state requiring 17 and 44 qubits at the two scales and 81 at $K=15$ while the register stays at eight throughout. Within the resolution of these experiments the parameter saving follows from that compression rather than from the expressive degrees of freedom of the circuit itself.

That saving carries a cost, and the regime in which it is repaid is narrow. Training the quantum actor costs roughly an order of magnitude more per episode than training its classical counterpart. The margin over the solver is bounded in a second sense. It is widest where the power budget is generous and reverses at 40 dBm and under the combined shift, the two operating points at which the solver itself ceases to reach full service, so the learned policies fall behind where the problem becomes constrained rather than uniformly.

VIII. LIMITATIONS

The quantum actor is evolved by state-vector simulation, so gate noise, the sampling error remaining beyond the shot budget modelled here, and compilation onto a device topology all lie outside the measurement, and the inference figures are algorithmic rather than hardware results. An eight-qubit register is within reach of present devices, but the $\approx 3.9 \cdot 10^4$ circuit evaluations each training episode consumes would be device time rather than simulated time, and it is that budget, not the register width, that separates the framework from execution on physical hardware.

The parameter ratio depends on where the boundary between the quantum and classical parts is drawn, falling to

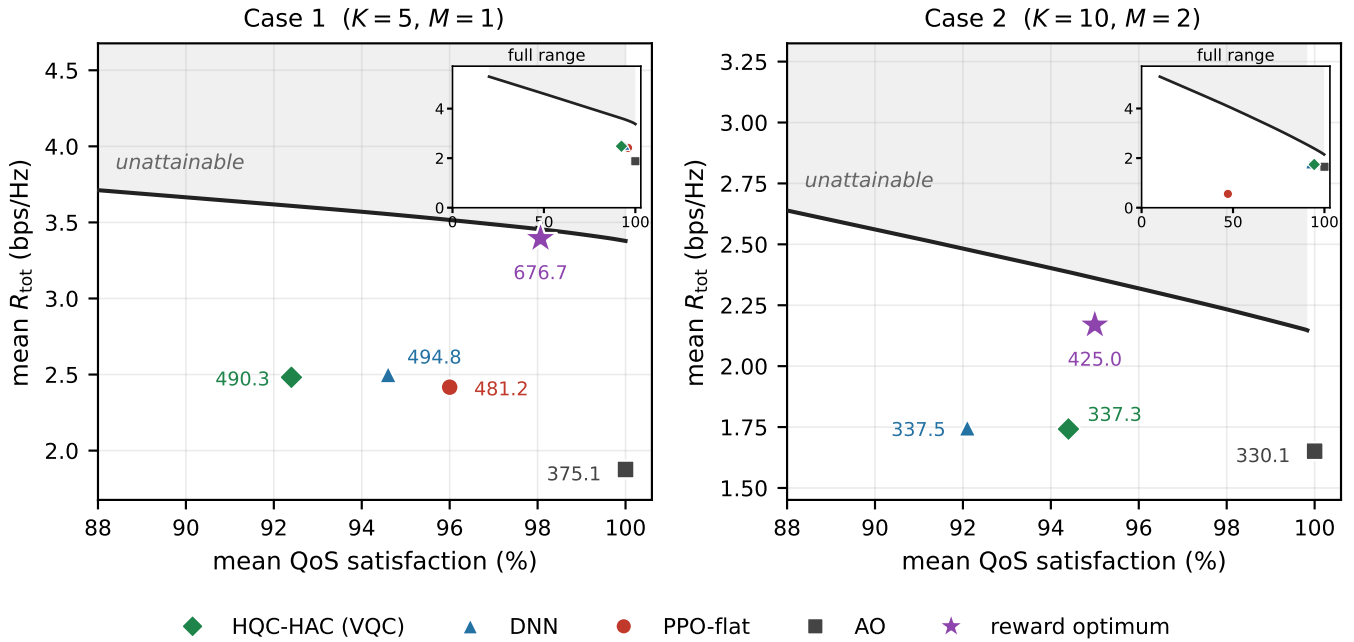


Fig. 6: Rate-QoS frontier and policy operating points for 600 test states per case. The frontier in (22) is obtained by a Lagrangian sweep over the achievable sets; since assignment is exhaustively enumerated and phase is solved exactly, while power splitting is grid-swept, it provides an achievable lower bound, with the shaded region unattainable under this construction. Markers denote the greedy-inference operating points from Table III, annotated by episode return, while stars indicate the reward-optimal operating points over the same achievable sets. The inset shows the full QoS range for Case 2, including PPO-flat.

roughly $1.4\times$ once the classical sub-actors the two arms share are counted. Every learned row averages five training seeds, at which resolution the QoS separation at (10, 2) is the sole difference between the two instantiations exceeding two standard errors.

The system model is restricted in two further respects. A single-antenna satellite admits only scalar precoding, so the spatial degrees of freedom of a multibeam payload, and the inter-group interference management they would afford, lie outside the formulation. The policy is also trained for a fixed user population and panel count, and the study spans two network scales and one channel family, so the narrowing of the margin over the solver with scale is observed rather than established.

IX. CONCLUSION

We presented HQC-HAC, a hybrid quantum-classical hierarchical actor-critic for sum-rate maximization in IRS-assisted grouped-RSMA integrated satellite-terrestrial networks. The formulation casts the joint assignment, phase, power, and common-rate problem as a sequential dependency chain under QoS, discrete-phase, blockage, and imperfect-CSI constraints. A variational quantum circuit resolves the combinatorial $(M+1)^K$ assignment, three classical sub-actors resolve the factorable low-level controls, and an autoencoder compresses the state so that the register size follows the design rather than the network size. Across two scales the method matches a classical actor of identical role, input, and output to within the spread across training seeds while using $9.2\times$ and $6.0\times$ fewer

parameters in the actor the two share, and it answers each channel realization with three evaluations of the rate kernel rather than the several hundred a per-state solver requires. It exceeds that solver by 31% at the smaller scale, the margin narrowing to within one standard deviation at the larger one. Future work includes execution on quantum hardware, a multibeam payload, a user population that varies without retraining, and a systematic sweep of circuit depth against classical width at matched budgets.

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